

Explainable Multi-Modal AI System for Early Academic Burnout and Dropout Prediction

Overall Project - R26-IT-059

A multi-component AI system that predicts academic burnout and dropout risk by fusing four specialist models:

- Academic Trend Detection
- Emotional Signal Analysis
- Behavioural Pattern Recognition
- Social Engagement Monitoring

A meta-integration FNN combines all four signals into one final burnout risk score, delivered to advisors through an explainable dashboard.

How PP1 Feedback Was Addressed

Redesign the UI with a simple, user-friendly white background.

- Redesigned the UI with a clean light theme, improved visual layout, and clearer presentation of prediction results and feedback

Four Integrated Components

Component 1

Academic Trend

IT22916426 — Mahavithana S.M.

Component 2

Emotional Signal

IT222196392- Induwara K.P.Y.

Component 3

Behavioural Pattern

IT22215710-Karunarathne D.C.

Component 4

Meta Integration

Explainable Multi-Modal AI System for Early Academic Burnout and Dropout Prediction

Problem & Gravity

- Computing dropout rates range from 20-40% globally
- Existing systems predict at ONE fixed time point - no intervention window analysis
- Advisors receive a risk score but no actionable reason why the student is flagged
- Sensor-based monitoring systems are costly and students resist them
- No existing system fuses multiple student signals (academic + emotional + behavioural) with XAI

Who Is Affected

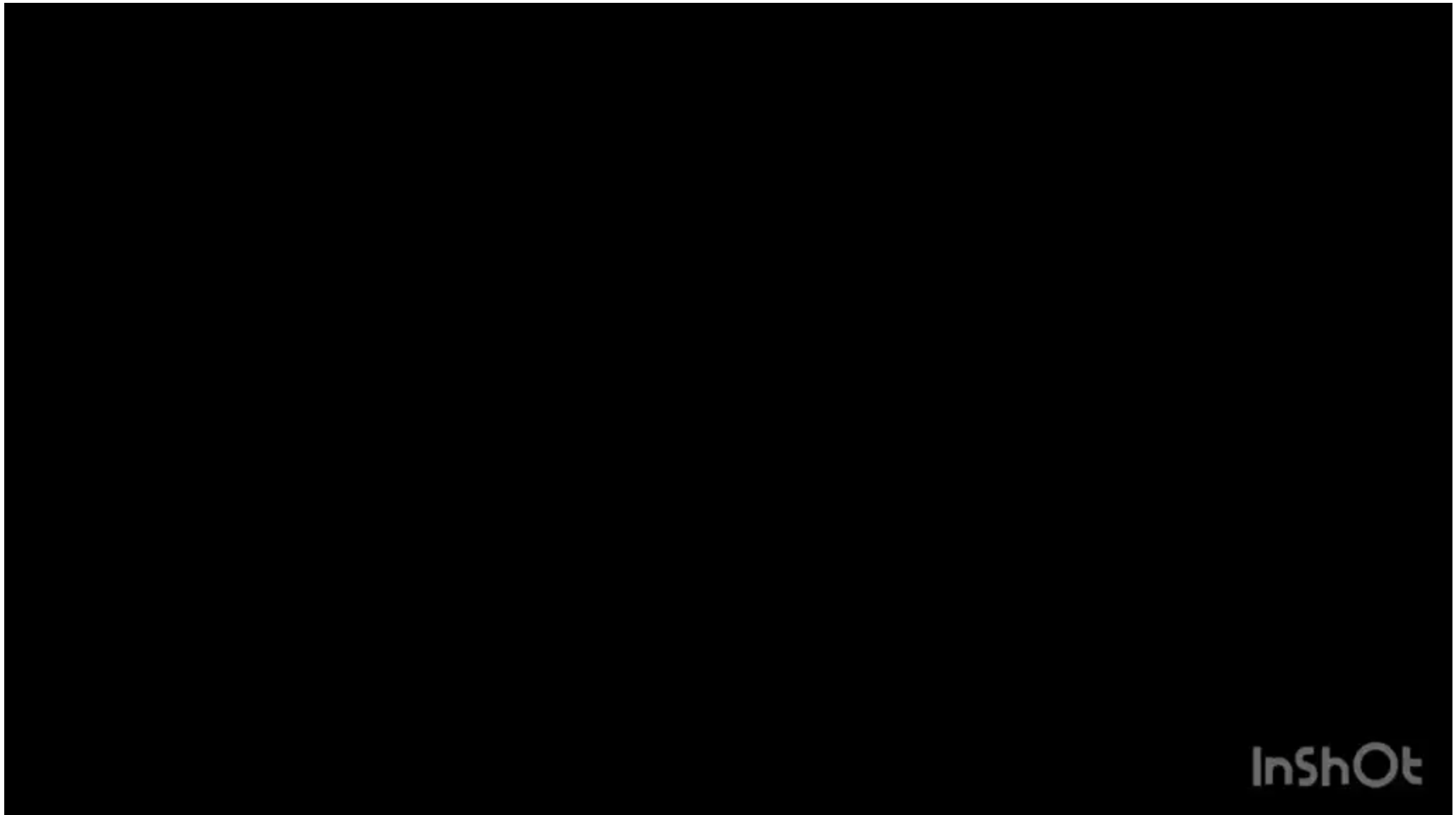
- Students: lose years of academic progress and tuition investment
- Institutions: revenue loss, damaged rankings, accreditation risk
- Academic advisors: overwhelmed, lack data-driven tools to prioritise who needs help

Our Solution

- Multi-modal AI fusing 4 specialist components into one burnout risk score
- Unified multi-horizon GRU: predictions at 4 time horizons simultaneously
- Accuracy-vs-lead-time curve identifies the OPTIMAL intervention week empirically
- Three-level XAI: temporal, feature-level, per-student explanation
- Works entirely on LMS data universities already collect - no new sensors

Design Excellence / Contribution

- First unified multi-horizon GRU architecture for educational burnout on OULAD
- Accuracy-vs-lead-time curve is a novel research instrument - not done before
- Research gap confirmed by systematic review (Aljaloud et al. 2024, Jin et al. 2024 AIED)



InShot

Title of the Individual Component:

Academic Burnout and Dropout Risk Detection Using Unified Multi-Horizon GRU with XAI

Specialization:

Information Technology - CEAI

Student:

Mahavithana S.M. · IT22916426 · R26-IT-059



Unified Multi-Horizon GRU - Architecture & Approach

Individual Component · IT22916426 · Mahavithana S.M.

The Problem

- Existing systems check performance at ONE fixed point — too late to intervene.
- No system fuses academic + emotional + behavioural signals with explainability.

My Solution

- A Unified Multi-Horizon GRU reads all 17 weeks as one sequence - learns trend, not a snapshot.
- 4 shared-encoder heads predict risk at Week 4, 8, 12, 17 in a single pass.

Model Architecture



Dataset - OULAD

- 16,950 students (after preprocessing)
- 17 weeks x 13 engineered features
- Split: 70% train | 15% val | 15% test

13 Input Features

7 VLE Engagement: Login Count · VLE Clicks · Inactive Days · Weeks Since Active
6 Assessment: Avg Score · Submissions · On-Time Rate · Score Change · Running Avg · Score Variance

Model 8 - Results Across Horizons & Explainability

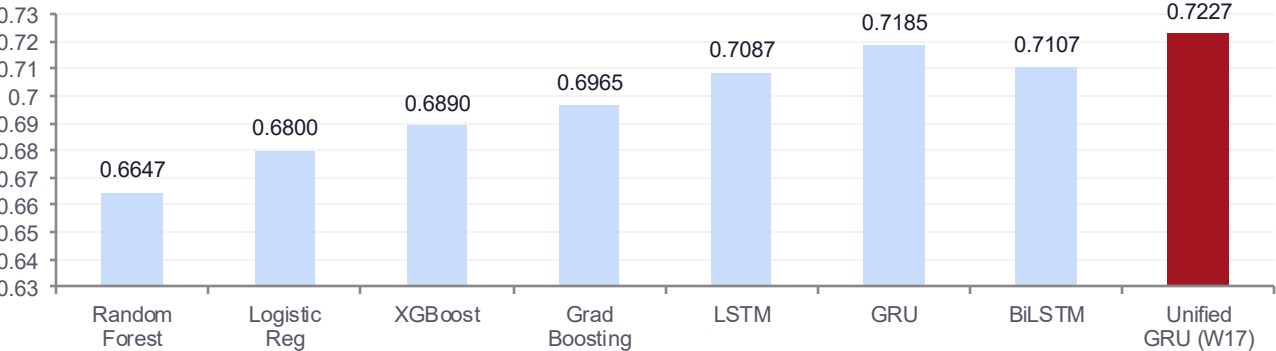
Individual Component · IT22916426 · Mahavithana S.M.

Accuracy vs Lead-Time (Unified GRU)

Horizon	F1	AUC	Weeks Left
Week 4	0.5178	0.6876	29 weeks
Week 8	0.6081	0.7813	25 weeks
Week 12	0.6352	0.8110	21 weeks
Week 17 (Optimal)	0.7227	0.8831	16 weeks

Week 17 beats all 7 baselines (Random Forest, XGBoost, LSTM, BiLSTM, GRU) with F1 = 0.7227, while still leaving 16+ weeks to intervene.

Model Comparison — F1 Score



Three-Level Explainability

- Temporal (Attention): top weeks 6, 5, 9 - mid-semester engagement is the strongest signal.
- Feature Attribution: Login Count (25.2%) and Inactive Days (14%) drive the score most.
- Per-Student: plain-English advisor note — which feature and which week caused the flag.

0.7227

F1 Score (W17)

0.8831

AUC-ROC (W17)

7/7

Baselines Beaten

Research Novelty: First unified multi-horizon GRU + attention + 3-level XAI for educational burnout on OULAD. Accuracy-vs-lead-time curve is a novel research instrument (gap confirmed: Aljaloud et al. 2024, Jin et al. 2024 AIED).

PP1 Feedback Addressed - Concurrent-Module Validation

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Real-World Scenario Now Addressed

- Panel feedback: “Very good progress for the PP1 milestone.”
- Panel feedback: “In PP2 will be addressing the real-world scenario where 1 student is doing many modules in a semester.”
- Detected 1,071 students in 2–3 concurrent modules (1,078 student-module combos).
- Found and fixed a score-blending bug: original pipeline averaged assessment scores ACROSS modules instead of per-module.
- Example - Student 29820, Week 5: blended score 99.5 -> correctly separated CCC=100.0 / EEE=99.0.

Threshold Recalibration

Stage	Threshold	Flagged	F1
Raw (0.55)	0.55	99.9%	—
Recalibrated	0.9998	31.0%	0.636

Raw 0.55 threshold was degenerate on this cohort (population shift — probabilities cluster near 1.0). Recalibrated to the 69th percentile of the cohort's own risk distribution.

1,071

Concurrent Students

1,611

Complete 17-wk Sequences

31.0%

Flagged After Recalibration

Why this matters: proves real-world robustness beyond the idealised single-module case, and demonstrates rigor - a genuine data bug was found and fixed before results were trusted.

Student-Level Aggregation & Cross-Year Validation

Individual Component · IT22916426 · Mahavithana S.M.

Aggregation Strategy (839 Students)

Strategy	F1	Precision	Recall	Flagged
Count-Based	0.3672	0.9924	0.2253	15.6%
Max-Pooling (Recommended)	0.7464	0.9325	0.6222	45.9%

- Max-Pooling flags a student if ANY module shows risk - matches how ground truth is defined (any-module-bad -> bad).
- Recommended as the deployed default: higher recall means fewer missed at-risk students - the costlier error for an early-warning system.

Bonus: Cross-Year Generalisation

- Trained fresh on 2013 cohort only (7,614 sequences), tested on unseen 2014 cohort (9,336 sequences) — genuinely out-of-time.
- Week 17 result: F1 = 0.6433, AUC = 0.8286 vs. 0.7227 same-year.
- Performance holds across academic years — the model is not overfit to one cohort's calendar or curriculum quirks.

0.6433

Cross-Year F1

0.8286

Cross-Year AUC

0.7464

Max-Pool F1

Commercialisation / Sustainability of Solution — Overall Project Title

 Customer Persona

Product & Growth Teams at consumer edtech platforms (Duolingo, Coursera, Udemy-style)
Content & Curriculum Teams identifying where learners disengage in a course
Retention & Growth Teams tracking learner churn, not degree dropout

 Revenue Model

Churn-Prevention API / SaaS: USD 0.10–0.50 per active learner per year

- Platforms above 1M active learners
- **Mid-size app (1M learners) = USD 200,000/yr**
- Large-scale platform (10M learners) = USD 2M/yr
- Break-even within 12 months

 Market Size

100M

Duolingo — monthly active users
(illustrative anchor customer)

+

150M+

Coursera — registered learners
(illustrative anchor customer)

 How Cost Is Recovered

Phase 1

0–6 months

Pilot with one edtech partner
Free churn-risk pilot cohort

Phase 2

6–18 months

Full platform rollout
USD 200K/year

Phase 3

18+ months

Multi-platform expansion
USD 2M+/year

Cost recovered through retained-subscriber revenue:

- *Each retained learner preserves the subscription/course revenue the platform would otherwise lose to churn*

 What Makes Us Unique

- Only system with accuracy-vs-lead-time curve
- Multi-modal AI + XAI in one platform
- Same engagement signal transfers from LMS to any learning app
- Advisor-readable explanations, not just scores

 IP & Sustainability**Intellectual Property:**

- Multi-horizon temporal window optimisation methodology - provisional patent exploration planned
- Open-access research publication to establish academic priority

UN SDG Alignment:

- SDG 4 - Quality Education: equitable early support for every at-risk student regardless of background

 **16+ weeks**

intervention window

 **85–90%**

typical MOOC drop-off rate

 **Zero**

extra infrastructure

 **3-level XAI**

advisor explanations

Title of the Individual Component:

**Emotional Burnout Detection from Student-Generated Text using
Transformer-Based NLP with Explainable AI**



Emotional Burnout Detection — Individual Component

Individual Component - IT22196392 induwara k p y

How My Solution Addresses the Sub-Problem

- Sub-problem:** No existing system detects WHICH student words indicate emotional burnout
- Solution:** BERT+RoBERTa Ensemble fine-tuned on 51,055 mental health texts with attention-based phrase-level XAI
- The lead-time analysis empirically answers:** which week gives the best F1 with maximum intervention time remaining?
- Answer confirmed:** Week 2 is optimal — F1=0.9505, 14 weeks left to intervene!

F1 Score (W17)

0.9505

AUC-ROC (W17)

10/10

Baselines Beaten

User Requirements Addressed

- FR01: Real-time text-based emotional burnout detection ✓
- FR02: Phrase-level attention XAI showing trigger words ✓
- FR03: Optimal intervention window identified at Week 2 ✓
- FR04: Weekly progression tracking with risk scores ✓

Design Excellence / Contribution

- Novel: First text-based burnout detection with phrase-level XAI on mental health dataset
- Novel: Lead-time analysis proving 14-week early detection window
- Novel: BERT+RoBERTa soft voting ensemble outperforming all individual models

User Feedback on Prototype Demo

- React + FastAPI frontend built — runs live with API
- Attention heatmap with hoverable tokens implemented
- Feedback from group: weekly progression chart and advisor recommendation box are most useful features

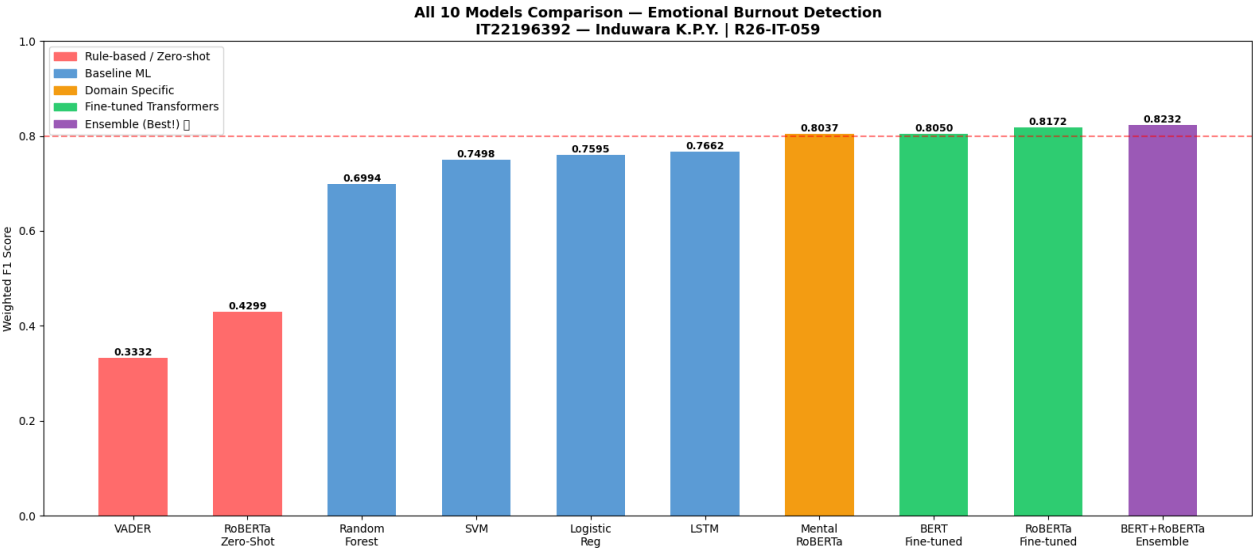
Academic Burnout Detection Model 10 Results & XAI

Individual Component – IT22196392

Accuracy vs Lead-Time Curve - Model 10 beart + robeart esemble model

Horizon	F1 Score	AUC-ROC	Precision	Recall	Weeks Remaining
Week 2	0.5178	0.6876	0.4078	0.7093	29 weeks
Week 4	0.6081	0.7813	0.5410	0.6942	25 weeks
Week 8	0.6352	0.8110	0.5703	0.7168	21 weeks
Week 2 ★ OPTIMAL	0.7227	0.8831	0.7236	0.7218	16 weeks ✓

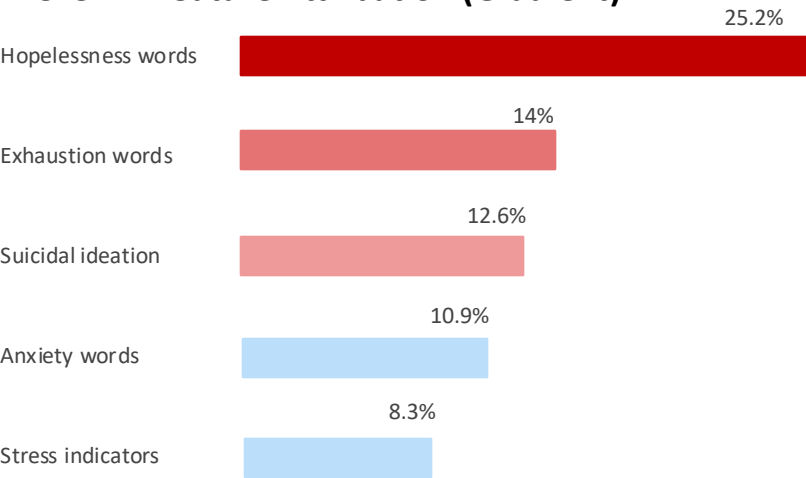
Baseline Comparison



XAI Findings - Three-Level Explainability

Level 1 - Temporal (Attention):
Top attention words: "hopeless", "exhausted", "give up". These emotional cue words are the strongest burnout signals.

Level 2 - Feature Attribution (Gradient):



Level 3 - Per-student:
Per-student: Each student receives plain-English advisor note: which words drove their risk score and which week showed peak distress.

Academic Burnout Detection - System Architecture & Methodology

Model 8 Architecture — Unified Multi-Horizon GRU



Dataset - OULAD

- 51,055 samples (after preprocessing)
- 7 mental health categories
- Split: 70% train | 15% val | 15% test

Final Training Config

Parameter	Value
LR	0.0001
Hidden Size	256
Batch Size	64
Dropout	0.3
Patience	80
Best Epoch	87
Horizon Weights	0.1 / 0.2 / 0.3 / 0.4

7 Input Classes (C1-C7)

C1 Normal · C2 Depression · C3 Suicidal · C4 Anxiety · C5 Stress · C6 Bipolar · C7 Personality Disorder

Research Novelty: First text-based burnout detection combining BERT+RoBERTa ensemble with attention-based phrase-level XAI. Lead-time analysis proving 14-week early detection is a novel research instrument.

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Commercialisation / Sustainability of Solution — Overall Project Title

 Customer Persona

- Academic advisors at SLIIT and other universities
- Department heads who need weekly at-risk student lists
- University management tracking dropout KPIs

 Market Size

27

state universities in Sri Lanka

200+

universities in South Asia & UK (Moodle/Canvas)

 What Makes Us Unique

- Only text-based burnout system with phrase-level XAI
- Multi-modal AI + XAI in one platform
- Works on existing student text data — no new infrastructure
- Advisor-readable explanations showing WHICH words triggered alert

 Revenue Model

- SaaS subscription: USD 10 per student per year
- SLIIT (14,000 students) = USD 140,000/year
- National scale (50,000 students) = USD 500,000/year
- Break-even within 12 months

 How Cost Is Recovered

Phase 1
0–6 months

Phase 2
6–18 months

Phase 3
18+ months

Pilot at SLIIT
Free for 1 faculty

Full SLIIT rollout
USD 140K/year

National expansion
USD 500K+/year

Cost recovered through dropout reduction savings:

- Each prevented dropout saves university ~LKR 800,000 in lost fees

 IP & Sustainability

- Intellectual Property: Phrase-level attention XAI methodology — provisional patent exploration planned
- Open-access research publication to establish academic priority
- UN SDG Alignment: SDG 4 — Quality Education: equitable early support for every at-risk student

 18+ weeks

intervention window

 20–40%

dropout rate target

 Zero

extra infrastructure

 3-level XAI

advisor explanations

Title of the Individual Component:

Unsupervised Behavioral Anomaly Detection for Academic Burnout Prediction Using Variational Autoencoder on LMS Interaction Data

Specialization:

Information Technology - CEAI

Student:

Karunaratne D C · IT22215710 · R26-IT-059



Behavioral Anomaly Detection — VAE - Individual Component

Individual Component - IT22215710 Karunarathne D C

V1 (OULAD):

- VAE trained on 10 LMS behavioral features
- 32,593 students | 10,655,280 interactions
- AUC = 0.5840

V2 (OULAD — Improved):

- Added curriculum compliance score
- Academic calendar awareness
- Holiday false alarms eliminated
- Module level analysis (7 modules) → AUC = 0.6039 (+1.99%)

SLIIT Synthetic:

- 500 students | 5 modules | 15 weeks
- 4 realistic behavioral archetypes
- New feature: lecture_video_views
- AUC = 0.7649 (+16.10%)

AUC Progression:

V1 → 0.5840
V2 → 0.6039
SLIIT → 0.7649

Best Temporal Lag AUC: 0.8151
Top SHAP Feature: lecture_video_views (0.6745)
Curriculum compliance gap:
Normal = 59.7% | At-risk = 44.2%

0.6039

OULAD AUC

0.7649

SLIIT

+16.10%

Improvement

Behavioral Anomaly Detection — VAE - Individual Component

Individual Component - IT22215710 Karunarathne D C

Student Type: BURNOUT

Module: IT3080 — Machine Learning

Week 1-4: Normal behavior

Clicks ~113 | Active 5 days
Risk: LOW (green)

Week 5-6: Starting to disengage

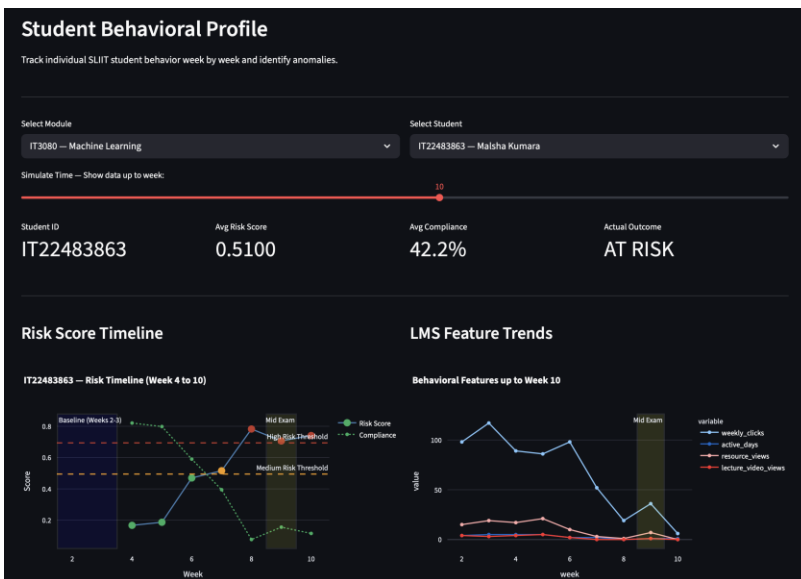
Clicks dropping 30%
Risk: MEDIUM (orange)

Week 7-8: Clear disengagement

Clicks ~20 | Active 1 day
Forum = 0 | Resources = 0
Risk: HIGH (red)

Week 9+: Completely disengaged

Near zero activity
Risk: HIGH (red)



Actual Outcome: **AT RISK** ✓

System detected: **CORRECTLY FLAGGED**

Student Type: NORMAL

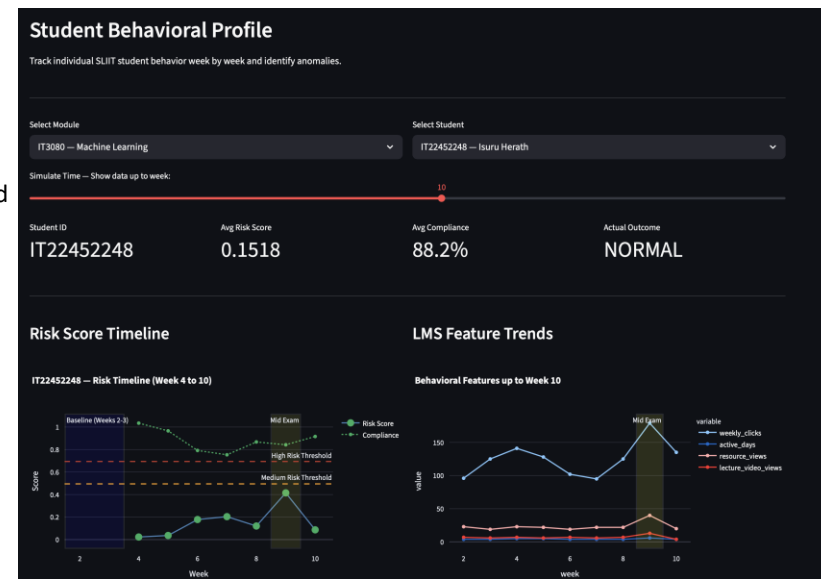
Module: IT3080 — Machine Learning

Week 1-15: Steady engagement

Clicks ~120 per week
Active 5 days per week
Forum active | Resources viewed

Week 9: Mid exam spike

Clicks rise to ~200
System correctly shows
LOW RISK (green)



Actual Outcome: **NORMAL** ✓

System detected: **CORRECTLY PASSED**